Reading map — where to find each week in the textbooks

You asked whether the module points you to the right chapter and page in Lehninger, Shuler and the others. The honest answer is: partly, and not to pages. I checked: of the 81 “what to open a book for” boxes in parts 2 to 14, 29 name a chapter number and none names a page. This map fixes what can be fixed.

What this map is built from, and why it has no page numbers

The chapter and section pointers come from your own curriculum, not from my memory. It names the editions — Lehninger 8th and Alberts 7th — and for weeks 1 to 8 it gives a chapter and the section to read in each session, such as “Lehninger Ch. 14, glycolysis through the aldolase step”. Those are more precise than anything I could add, and they are checkable against your copy, so I have used them verbatim in the third column.

Page numbers I cannot give. They change between editions and printings, I have no access to the books, and a wrong page quoted confidently sends you to the wrong place and costs more time than a pointer to a chapter. So the last column is blank: fill in the page in *your* copy the first time you look it up, and the map becomes your own index. For the engineering books — Shuler, Doran, Stephanopoulos, Cornish-Bowden — the curriculum names the topic (“the enzyme-kinetics chapter”) and not a number, and I have kept to that; chapter numbers differ between editions of those books, and I would rather you check the contents page than trust a number I recalled.

The “look up in the index” column is mine. They are the terms to search in a book’s back-of-book index. Suggestions, not citations.

Module location: part 01 and part 14 both cover weeks 1 to 10 (part 14 is the compiled core concepts); S1 to S6 are the supplementary chapters in part 13. File names are in the index.

A — Weeks with a textbook or reading pointer

Wk	In the module	Textbook pointers from your curriculum	Look up in the index	Your page
1	part 01 · ch 1 part 14 · ch B1	· Lehninger Ch. 1, the cellular-foundations section · Lehninger Ch. 1, the rest of the chapter: chemical, physical, genetic and evolutionary foundations · Alberts Ch. 1, skim; then read the central-dogma pages carefully · Alberts Ch. 2, the sections on membranes and organelles	central dogma · prokaryotic cell · macromolecular composition · genetic code	
2	part 01 · ch 2 part 14 · ch B2	· Lehninger Ch. 3, amino acids and peptide bonds · Lehninger Ch. 4, the four levels of structure · Lehninger Ch. 6, the sections before kinetics: what enzymes do and how they lower activation · Lehninger Ch. 6, the cofactor and coenzyme material	amino acids, ionisation · isoelectric point · peptide bond · Ramachandran plot · protein folding · coenzymes, vitamins	
3	part 01 · ch 3 part 14 · ch B3 part 13 · S3	· Lehninger Ch. 6, the enzyme-kinetics section · Lehninger Ch. 6, the regulatory-enzymes section	Michaelis–Menten · Briggs–Haldane · turnover number · inhibition, competitive · Hill equation · allosteric enzymes	
4	part 01 · ch 4 part 14 · ch B4	· Lehninger Ch. 13, the bioenergetics and thermodynamics section · Lehninger Ch. 13, phosphoryl-group transfers and ATP · Lehninger Ch. 13, the biological oxidation–reduction section	free energy, standard · phosphoryl group transfer · ATP hydrolysis · reduction potential · NAD/NADP	
5	part 01 · ch 5 part 14 · ch B5	· Lehninger Ch. 14, glycolysis through the aldolase step · Lehninger Ch. 14, the payoff phase to pyruvate · Lehninger Ch. 14, the pentose phosphate pathway section · Lehninger Ch. 15, principles of metabolic regulation — the sections on glycolytic control	glycolysis, regulation · phosphofructokinase · pentose phosphate pathway · Entner–Doudoroff · phosphotransferase system	
6	part 01 · ch 6 part 14 · ch B6	· Lehninger Ch. 16, the pyruvate dehydrogenase complex · Lehninger Ch. 16, the cycle itself · Lehninger Ch. 16, the anaplerotic reactions section · Lehninger Ch. 19, through ATP synthase	citric acid cycle · pyruvate dehydrogenase complex · anaplerotic reactions · glyoxylate cycle · chemiosmotic coupling	
7	part 01 · ch 7 part 14 · ch B7	· Lehninger Ch. 22, nitrogen assimilation and the glutamate–glutamine node · Lehninger Ch. 21, fatty-acid synthesis · Read the mevalonate pathway in Lehninger Ch. 21 and in KEGG map 00900 · Lehninger Ch. 22, nucleotide biosynthesis, read for the shape rather than every intermediate	fatty acid synthesis · amino acid biosynthesis · glutamine synthetase · mevalonate pathway · nucleotide biosynthesis	
8	part 01 · ch 8 part 14 · ch B8	· Alberts Ch. 5 on replication, then Ch. 6 on transcription · Alberts Ch. 6, translation · Alberts Ch. 7, transcriptional control; then read the lac operon specifically	DNA replication, bacterial · RNA polymerase, sigma factor · Shine–Dalgarno · lac operon · translation initiation	
9	part 01 · ch 9 part 14 · ch B9 part 13 · S4	no chapter assigned; see the tracker	catabolite repression · cAMP receptor protein · stringent response · two-component systems · quorum sensing	
10	part 01 · ch 10 part 14 · ch B10	· Madigan et al. — Brock Biology of Microorganisms, the chapters on microbial growth, diversity and genetics	growth curve · Monod kinetics · chemostat · mutation, spontaneous ·	

		· Snyder & Champness — Molecular Genetics of Bacteria, on selections, transformation and conjugation	Luria–Delbrück · conjugation · plasmid copy number · sterilisation	
11	part 02 · ch 11	· Bailey, “Toward a science of metabolic engineering”, Science 252:1668 (1991) · Stephanopoulos & Vallino, “Network rigidity and metabolic engineering in metabolite overproduction”, Science 252:1675 (… · Nielsen & Keasling, “Engineering cellular metabolism”, Cell 164:1185 (2016)		
13	part 02 · ch 13	· Stephanopoulos Ch. 8, network stoichiometry · Stephanopoulos Ch. 12, determined and underdetermined systems · Stephanopoulos Ch. 13, pathway structure; or the corresponding Palsson chapters	stoichiometric matrix · null space · degrees of freedom · flux balance analysis · elementary flux modes	
14	part 02 · ch 14	· Lehninger Ch. 13, reread the thermodynamics section — this time for ΔG′ rather than ΔG°′ · Lehninger Ch. 11, the passive-transport sections · Lehninger Ch. 11, the active-transport sections · Stephanopoulos Ch. 11, metabolic control analysis, the first half · Stephanopoulos Ch. 11, the rest	free energy of reaction · mass-action ratio · membrane transport · proton antiporter · weak acid, pH partitioning	
15	part 02 · ch 15	no chapter assigned; see the tracker	metabolic control analysis · flux control coefficient · summation theorem · elasticity	
16	part 02 · ch 16 part 13 · S6	· Stephanopoulos, Aristidou & Nielsen, the chapters on cellular-reaction kinetics	kinetic modelling · Haldane relationship · parameter estimation · identifiability	
17	part 02 · ch 17	no chapter assigned; see the tracker	statistical power · multiple comparisons · factorial design · replication	
18	part 02 · ch 18	· Doran — Bioprocess Engineering Principles, the material- and energy-balance chapters	yield coefficient · material balance · theoretical yield · energy balance	
19	part 02 · ch 19	no chapter assigned; see the tracker	NADPH supply · transhydrogenase · oxygen demand · P/O ratio	
23	part 07 · ch 23	· Snyder & Champness — Molecular Genetics of Bacteria, on recombination, selection and counterselection		
30	part 08 · ch 30	· Cornish-Bowden — Fundamentals of Enzyme Kinetics, for what a kinetic constant does and does not tell you	enzyme kinetic constants · k_cat/K_M · control coefficient	
35	part 10 · ch 35	· Install COBRApy and a solver; confirm the install by loading the textbook model · Read Orth, Thiele & Palsson, “What is flux balance analysis?”, Nat. Biotechnol. 28:245		
37	part 10 · ch 37	· Read the Thiele and Palsson protocol for what manual reconstruction actually involves — it is a year		
43	part 11 · ch 43 part 13 · S5	· Doran — Bioprocess Engineering Principles, 2nd ed., the mass-transfer and reactor chapters, and work its problems · Shuler, Kargi & DeLisa — Bioprocess Engineering, the chapters on reactor operation and scale-up	isotopic tracers · mass isotopomer distribution · metabolic flux analysis	
44	part 11 · ch 44 part 13 · S1, S6	no chapter assigned; see the tracker	oxygen transfer · k_L a · power consumption · mixing time · fed-batch · scale-up	
49	part 04 · ch 49 part 13 · S1	no chapter assigned; see the tracker	bioseparations · downstream processing cost · process economics	
59	part 06 · ch 59	· Madigan et al. — Brock Biology of Microorganisms, the cell-envelope chapter	cell envelope, Gram-positive · Corynebacterium	
60	part 06 · ch 60	· Madigan et al. — Brock Biology of Microorganisms, the phototrophy and microbial diversity chapters	phototrophy · cyanobacteria · microbial diversity	

B — Weeks whose sources are papers, reviews and tools

These weeks have no textbook in your curriculum; the first two sources it lists are shown. The full list is on each week’s page of the tracker.

Wk	In the module	First sources named in your curriculum
12	part 02 · ch 12	· NumPy — the absolute basics and linear algebra guides · SciPy — integration and optimization reference
20	part 07 · ch 20	· Addgene, “Plasmids 101: Golden Gate cloning” · Gibson et al., Nat. Methods 6:343 (2009) tooling, Golden Gate design, and yeast in vivo assemb…
21	part 07 · ch 21	see the week’s sources list in the tracker
22	part 07 · ch 22	see the week’s sources list in the tracker
24	part 07 · ch 24	· Cleto et al., “C. glutamicum metabolic engineering with CRISPR interference”, ACS Synth. Biol.… · A metabolite-biosensor review
25	part 07 · ch 25	· Fredens et al., Total synthesis of Escherichia coli with a recoded genome, Nature (2019) · Weinstock et al., Vibrio natriegens as a fast-growing host for molecular biology (preprint, 20…
26	part 08 · ch 26 part 13 · S4	· A review of transcription-factor-based biosensors for metabolic engineering · Kelly et al., Measuring the activity of BioBrick promoters using an in vivo reference standard…
27	part 08 · ch 27 part 13 · S4, S6	· Yoo et al., Metabolic engineering of Escherichia coli using synthetic small regulatory RNAs, N… · CRISPR interference-guided multiplex repression of endogenous competing pathway genes for redi…

28	part 08 · ch 28	· Salis Lab RBS Calculator · MIT OCW 7.344 Directed Evolution: Engineering Biocatalysts
29	part 08 · ch 29	· Dauparas et al., Atomic context-conditioned protein sequence design using LigandMPNN, Nat. Met… · Institute for Protein Design — introducing LigandMPNN
31	part 08 · ch 31	see the week's sources list in the tracker
32	part 09 · ch 32	· breseq — documentation and downloads · breseq on GitHub
33	part 09 · ch 33	· Love, Huber & Anders, Moderated estimation of fold change and dispersion for RNA-seq data with… · Bioconductor — DESeq2 package and vignette
34	part 09 · ch 34	· K-Biofoundry · Global Biofoundry Alliance
36	part 10 · ch 36	· COBRApy documentation — building a model, simulating, deletions, FVA · COBRApy — flux analysis guide
38	part 10 · ch 38	· The COBRA Toolbox — documentation and installation · MATLAB — getting started documentation
39	part 11 · ch 39	· Burgard, Pharkya & Maranas, "OptKnock", Biotechnol. Bioeng. 84:647 (2003) · A ^{13}C-MFA methods review
40	part 11 · ch 40	· Delépine et al., "RetroPath2.0", Metabolic Engineering 45:158 (2018) · A 2025 review on machine learning or AI-powered biofoundries
41	part 11 · ch 41	· Radivojević et al., A machine learning Automated Recommendation Tool for synthetic biology, Na… · Zhang et al., Combining mechanistic and machine learning models for predictive engineering and…
42	part 11 · ch 42	· PubMed search: machine-learning-guided directed evolution of proteins · PubMed search: protein language models and fitness prediction
45	part 03 · ch 45	· Dunlop et al., Engineering microbial biofuel tolerance and export using efflux pumps, Mol. Sys… · Newman et al., High-level production of amorpha-4,11-diene in a two-phase partitioning bioreac…
46	part 03 · ch 46	· Prpich and Daugulis, Polymer development for enhanced delivery of phenol in a solid–liquid two… · Rehmann, Sun and Daugulis, Polymer selection for biphenyl degradation in a solid–liquid two-ph…
47	part 03 · ch 47	· Newman et al., High-level production of amorpha-4,11-diene in a two-phase partitioning bioreac… · Substrate mass transport in two-phase partitioning bioreactors employing liquid and solid non-…
48	part 03 · ch 48	· PubMed search: cell immobilisation in alginate and PVA, reviews of methods and limitations · PubMed search: effectiveness factor and oxygen diffusion limitation in immobilised-cell beads
50	part 04 · ch 50	· A tutorial on analytical method validation for chromatographic assays · A review of quenching and extraction for microbial metabolomics
51	part 04 · ch 51 part 13 · S5	· Ko, Lee et al., "Tools and strategies of systems metabolic engineering", Chem. Soc. Rev. 49:46… · A gas-fermentation or scale-up review
52	part 04 · ch 52	· REBASE — the restriction enzyme database · Wetmore et al., Rapid quantification of mutant fitness in diverse bacteria by sequencing rando…
53	part 05 · ch 53	· Paddon et al., Nature 496:528 (2013) · Paddon & Keasling, Nat. Rev. Microbiol. 12:355 (2014)
54	part 05 · ch 54	· Yim et al., Nat. Chem. Biol. 7:445 (2011) · EPA, 2003 Greener Reaction Conditions Award — DuPont bio-PDO
55	part 05 · ch 55 part 13 · S2	see the week's sources list in the tracker
56	part 05 · ch 56 part 13 · S2	· AIM Lab — KIST Applied and Industrial Microbiology Lab (publications page) · Sun et al., efficient poly(3-hydroxybutyrate) production from lignocellulosic hydrolysates wit…
57	part 05 · ch 57 part 13 · S2	· AIM Lab — KIST Applied and Industrial Microbiology Lab (home) · Firmansyah et al., Upgrading polyethylene plastic waste into a biodegradable polymer: a hybrid…
58	part 06 · ch 58	see the week's sources list in the tracker
61	part 06 · ch 61	· A 2025–2026 review on AI-driven biofoundries or systems metabolic engineering · A cell-free metabolic engineering review
62	part 12 · ch 62	· ICMJE recommendations on authorship and contributorship · COPE, Committee on Publication Ethics: guidance and flowcharts
63	part 12 · ch 63	· Mensh & Kording, Ten simple rules for structuring papers, PLoS Comput. Biol. (2017) · A guide to presenting scientific data and figure design
64	part 12 · ch 64	see the week's sources list in the tracker
65	part 12 · ch 65	· Your own paper log entries · Two recent papers from each of your three target labs
66	part 12 · ch 66	see the week's sources list in the tracker

One habit that makes this work

When a module chapter says "open a book for this", find that week in table A, read the pointer in your curriculum for the section, and write the page in your copy into the last column. After ten weeks you have a page-level index of your own editions, which is the thing I cannot write for you.